"Real-Time Object Detection Using ML-Powered Geolocation Tracking"

[1] Vishwanjali k
Dept. of Computer Science and Engineering
Jain University
Bangalore, India
Vishwanjalikoppula@gmail.com

[2] Hanisha k
Dept. of Computer Science and Engineering
Jain University
Bangalore, India
hanishakottlo288@gmail.com

[3] Poojitha k
Dept. of Computer Science and Engineering
Jain University
Bangalore, India
Kurampoojitha@gmail.com

[4] Mythri Siri N
Dept. of Computer Science and Engineering
Jain University
Bangalore, India _
my3siri2005@gmail.com

[5] Ushodaya N
Dept. of Computer Science and Engineering
Jain University
Bangalore, India
ushodayanamala9@gmail.com

[6] Sai Rishika M
Dept. of Computer Science and Engineering
Jain University
Bangalore, India
mekalasairishika@gmail.com

***Abstract* — The geolocation tracking system would be a Wi-Fi based system that proposes to find professors within a university campus in real-time. The system employs the Wi-Fi triangulation and signal strength estimation used to estimate the positions indoors with the use of existing Wi-Fi access points. The system uses the values of the Received Signal Strength Indicator (RSSI) to compute precise location values in buildings with more than one storey. Machine learning algorithms are used to enhance the precision of positioning and forecast the movement patterns using the past data. The solution proposed does not require GPS or Bluetooth and, therefore, is inexpensive and can be installed easily with the existing campus systems. The system monitors attendance automation and improves security of the campuses with real-time tracking systems. A server-client design is adopted to process, store and visualize the location data effectively. Encryption, anonymization, and controlled access are the means to ensure data privacy. The techniques used in performance optimization are used to manage performance variations and interference. In general, the system offers a scalable, secure, and efficient smart campus tracking system with Wi-Fi.**

Keywords— Wi-Fi Triangulation, Indoor Positioning System (IPS), Geolocation Tracking, Machine Learning, RSSI (Received Signal Strength Indicator),Smart Campus, Attendance Automation, Movement Prediction, Data Privacy and Security.

I. INTRODUCTION

Monitoring of professors in a university college will be significant in enhancing the efficiency of administration and management of the campus. The conventional means of attendance like manual attendance register and sign in are time consuming and they are also subject to human errors. Such outmoded practices usually result in poor records and wasteful use of resources.

Due to the recent blistering development of wireless communication technologies, Wi-Fi has become one of the trusted solutions to indoor positioning. Wi-Fi infrastructure is accessible in most university campuses, which makes it an affordable and scaled way of real-time tracking. The Wi-Fi signal can be used effectively to give the exact location of an indoor position as opposed to GPS which cannot work well indoors.

The principle of Wi-Fi-based positioning is that it compares the Received Signal Strength Indicator (RSSI) values of two or more access points. These values of signal strength are worked out to give the approximate position of a device in a building. The system will be able to detect changes and can be trained to be more accurate by applying machine learning algorithms to the RSSI data.

The proposed system is based on the Wi-Fi technology completely to establish a professor locating system within a campus location. It utilizes the presence of accessible Wi-Fi access points to gather signal data without making any hardware installations. This renders the system cost effective and simple to adopt.

The Wi-Fi signal data obtained is sent to a central server where preprocessing and filtering processes are done. Historical data on signal is used to train machine learning models that map patterns on RSSI to particular locations. The system becomes skilled over time and learns movement patterns and becomes more accurate in predicting.

The main aim of this system is to automate the process of tracking professor attendance as well as giving real-time updates of locations to the administration aiming at providing real-time information. It is also able to improve security in the campus by detecting abnormal movement patterns. Also, the system facilitates effective planning and allocation of resources in the classroom.

The system design has a high priority towards data privacy and security. All the data through Wi-Fi is encrypted, anonymized, and stored safely to adhere to the ethical principles and laws of data protection.
Even though the Wi-Fi signals can vary in the case of interference and changes in the environment, due to advanced filtering and learning methods, the performance is not upset. Using the Wi-Fi infrastructure alone, the proposed system can offer a viable and sound scaling solution to smart campus management.
Altogether, this Wi-Fi based tracking system is a new step to creating an intelligent, secure and efficient academic environment.

II. RELATED WORK

The limitations of GPS in the indoor environment have been well researched and researchers have explored the use of indoor positioning to utilize available infrastructure. The Wi-Fi signals, which have already become ubiquitous on campuses, have been a favorite target since they can be used in positioning without incurring hardware expenses. The first systems based on Wi-Fi utilized fingerprinting, in which signal strengths at the reference points of various access points are measured, and a database is constructed. The RADAR system developed one of the earliest and practical Wi-Fi fingerprinting systems, which showed that Received Signal Strength Indicator (RSSI) patterns can be projected onto physical locations with decent precision.
Following studies came up with better methods of fingerprint matching based on pattern and machine learning. RSSI measurements were used with conventional classifiers like k-Nearest Neighbors (k-NN), Support Vector machines (SVM) and Decision Trees to allocate locations in similarity measures, proving to be stronger than simple look up methods. Likelihoods of location due to the presence of uncertainty in the signal have also been estimated by probabilistic approaches such as Bayesian inference and Gaussian mixture models, which are more resistant to noise and variability.
Due to the multipath effects and change of the environment, Wi-Fi signals have low stability, which is why numerous researchers have proposed signal preprocessing and filtering methods to enhance stability. Smoothing, normalization, and the removal of outliers are often used to minimize the effects of momentary spikes of the signal.
Recent activity has welcomed the use of machine learning in order to refine positioning even further. Neural networks, especially deep learning models such as Convolutional Neural Networks (CNNs), have been conditioned with RSSI fingerprints in order to automatically derive signal features and represent spatial associations, which are superior in a complicated indoor environment compared to conventional methods. They have also been captured using recurrent models like LSTMs that capture temporal trends of signal sequences as users traverse through spaces..
Also, the adaptive and online learning solutions have been suggested as the models where models receive new Wi-Fi data to adjust to the changes in the environment, including new access points or moved furniture that may have a substantial impact on the RSSI characteristics.
Localization of Wi-Fi in the smart campus application has facilitated services such as the navigation assistance, occupancy analytics and movement of crowds. These systems explain how Wi-Fi positioning can be used to serve more operation requirements than the mere estimation of the location.
Though a lot of research has been made, there are still challenges including heterogeneity of devices where various mobile devices measure RSSI differently and this is a problem when it comes to consistency of models. This has been addressed by the suggestion of calibration procedures and device-specific modeling.
Although various researches have been performed using Wi-Fi positioning either to track students or assets within educational institutions, a number of them concentrate on real-time tracking of faculty members that is accompanied by automated functions such as attendance tracking and movement forecasting. Most of the existing systems also use other sensors, inertial measurement sensor, to enhance accuracy.
The issue of privacy and ethical data management have been identified as areas of concern in the recent work and solutions such as encryption, anonymization, and access control are offered to protect the location data of the users.
The corresponding literature highlights the fact that Wi-Fi-only systems represent a convenient, affordable answer to the problem of indoor positioning and tracking. Nonetheless, a combination of machine learning to gain precision and customization of systems to real-time operational purposes, including the presence and attendance of professors, is a developing topic. This paper is based on these premises to construct a Wi-Fi based tracking infrastructure, that focuses on smart campus performance, scalability and privacy..

Table 1: Summary of critical evaluation

STUDY	METHODOLOGY
[1] Bahl and Padmanabhan (2000)- RADAR System Bakhsh	Wi-Fi Fingerprinting (RSSI) RSSI values at known points were collected and the real-time signal was compared with the stored database.

[2] Li et al. (2017) - Smart Campus Prediction	k-NN, SVM on Wi-Fi RSSI -Used machine learning models to classify location by signal patterns
[3] Campos & Garcia (2018)-Wi-Fi Indoor Navigation	RSSI-based Localization The Wi-Fi signal strength was measured and used to determine the position indoors.
[4] Gupta & Thomas (2021) - Anomaly Detection	ML-based RSSI Analysis
[5] Zhan et al. (2016) - Hybrid Positioning.	Wi-Fi RSSI Modeling

III. METHODOLOGY

The suggested system will be a Wi-Fi based indoors positioning system to provide the ability to track professors accurately and in real-time within a university campus. The technique exclusively uses the existing Wi-Fi infrastructure and does not require extra hardware to operate, including GPS devices or Bluetooth pads. The initial step will be a strict site survey of the campus buildings to locate and map all available Wi-Fi access points. The physical coordinates of each access point and its coverage area are recorded in order to create a reference framework.

.

After analysis of infrastructure, a data collection phase will be initiated in order to build a Wi-Fi fingerprint database. The values of RSSI (Received Signal Strength Indicator) are measured at various pre-specified points of reference areas including classrooms, laboratories, corridors, faculty cabins and seminar halls. A number of signal readings are recorded at the various reference locations as time progresses, and this is done to reduce short-term signal changes. The repeated measurements are used to enhance the accuracy and minimize the interferences and the multipath effect

Raw data is accumulated in a centralized storage database where further processing can be done. Preprocessing of the data is used in the improvement of signal quality and elimination of inconsistencies.

The filters to noise cancelation remove the fluctuating values due to interfering environments. The outlier detection algorithms detect and reject the abnormal signal values. Normalization of the signal is done to make the RSSI measurements of two or more devices homogenous. The process is followed by the feature extraction techniques to shortlist the most pertinent signal attributes that can be used to train the model.

Following the preprocessing, labels of the dataset based on known physical locations are assigned, and the dataset is further partitioned into the training and testing sets. The machine learning algorithms that are used include k-Nearest Neighbors (k-NN) and Support Vector Machines (SVM) because they are helpful in classification tasks. The training stage allows the model to acquire the correlation between the pattern of RSSI and the respective locations indoors. Hyperparameter tuning performs the optimization of the model parameters and predictive accuracy. The cross-validation methods are adopted to avoid overfitting and to guarantee model generalization.

The evaluation of model performance is based on accuracy, precision, recall, and confusion matrix, and the retraining of the model, when necessary, to achieve better results.

Once the trained model attains reasonable accuracy, it is put into service on a centralized server to serve in real-time.

The data on live RSSI are being continuously gathered, safely transmitted and analyzed to forecast where one is today.

A database holds the predicted locations to be used in reporting, analysis of movement and automated attendance.

The system is based on encryption, access control, recalibration, and adaptive learning, which guarantees secure, scalable, and efficient Wi-Fi-based smart campus tracking.

IV. PROPOSED ARCHITECTURE

Proposed Architecture:

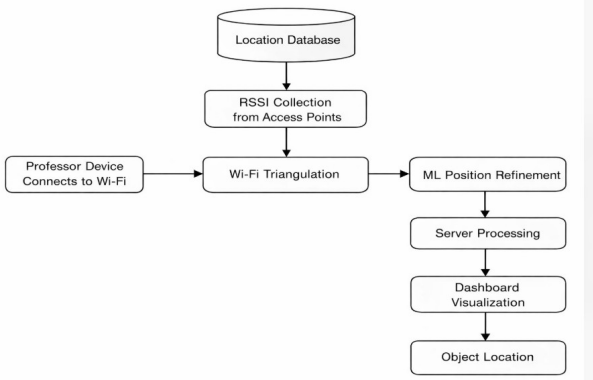

fig 1: Block diagram of the Real time Wifi system.

Figure 1-The proposed system will be initiated when a professor has his/her device connected to a Wi-Fi network of the university, and it is allowed to inter-relate with the numerous access points. The value of RSSI (Received Signal Strength Indicator) is constantly measured and inserted into the location database to be processed. Wi-Fi triangulation is done with the help of these signal strength measurements to determine the approximate position of the professor indoors. A machine learning model also helps to refine the estimated data to enhance the localization accuracy. The centralized server processes the processed results and presents them as a displayed dashboard interface with which real-time visualization is provided. According to the estimated

location, the system will automatically mark attendance or commence other needed security-related tasks, which facilitates smart and effective campus management

• **Location Database:**Location Database is the major reference system of the whole setup. It has information in details regarding the building structure such as the room numbers, the floor layout and coordinates of all Wi-Fi access points installed in the area which is used to compare real-time signal reading with known reference points which is the foundation of good indoor positioning.

• **RSSI Collection from Access Points:**
RSSI (Received Signal Strength Indicator) is used to determine the strength of a received wireless signal. Every wireless LAN access point has a sensor constantly monitoring the signal of the devices that are connected. When a device is operational in the network, we have several access points that have the values of RSSI at the same time. As the signal strength reduces with distance (and depends on walls, furniture, and interference), multiple distance-related indicators can be obtained by gathering RSSI data at different locations, which are necessary in determining position.

• **Professor Device Connects to Wi-Fi:**As the smartphone, tablet, or laptop of the professor is connected to the Wi-Fi network of the institution, it is visible in the system. The device carries a unique identifier in the network (like a MAC address or authenticated user id) which enables the system to identify it as one that belongs to a particular professor. The device constantly communicates with nearby access points as long as it is connected, making it possible to determine the change in signal strength in real time and maintain a consistent feature representation between devices

.

• **Wi-Fi Triangulation:**
Wi-Fi triangulation involves at least three values of RSSI of different access points in determining the physical location of the device. The system provides coverage circles around each router that overlap by estimating the distance of signals and converting these values into signal strength. The point of intersection of these circles gives an approximation of the location of the device. Simple triangulation will provide a rough location, but the accuracy will be low because of the variation of signals and the environment..

.

• **ML Positioning:** Triangulation is followed by application of machine learning algorithms to enhance the accuracy of positioning.

• They are trained with signal data on historical signals taken at known locations indoors.

The model does not use mathematical distance formulas, but learns complex signal patterns which include interference, reflections and noise.With time, it offers more precise room-level or zone-level location predictions than the conventional triangulation techniques.

• **Server Processing:** Once this location is estimated it is sent to a central server and bad data is discarded, the user is authenticated and the records are stored.

• The server can also be linked to the attendance or security systems to have automatic actions.

• The last point of interest is the position on a dashboard map, where administrators can monitor the position of the professor and monitor attendance with ease.

• **Security Action:**Lastly, the validated data of positions is applied to activate automatic system activities.

• In case a professor is caught in one of the planned classrooms, automatic attendance is registered and notifications are sent in case an individual enters the restricted premises.

• This automation saves on labor, manpower, as well as increases efficiency, safety and accountability to the institution.

Data Set training:

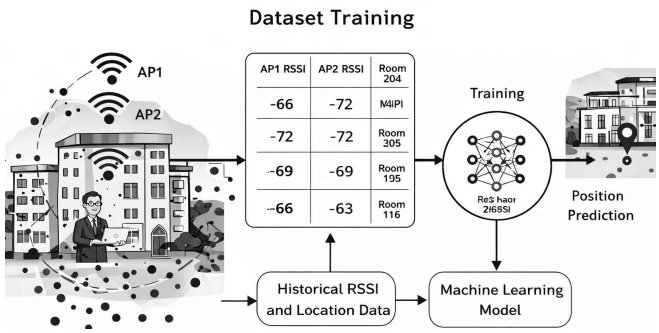

Fig 2: Wi-Fi RSSI Dataset Preparation and Model Training Architecture.

Project Implementation Steps

1. The initial stage, Infrastructure Setup, uses the currently available access points of the campus Wi-Fi to prevent the need to purchase additional hardware. The system is set up to receive Received Signal Weakness Indicator (RSSI) of the devices that will be linked to more than one access point. A safe server-Client architecture is created so that continuous communication can exist between devices of the users and the main processing server.

2. On the second stage, Localization Development, a Wi-Fi triangulation algorithm is adopted and it is used to estimate the indoor positions based on RSSI signal strength values. Positioning is enhanced by the use of signal propagation models. Special calibration systems are added in order to support multi-storey buildings and to counterattenuation of signals, interference, and environmental variations.

3. The third stage is the Machine Learning Integration. Machine learning models, which improve positioning accuracy, are trained using historical RSSI data and their respective location coordinates. Locations are refined by the ML algorithms, and predictive analytics is made possible to predict movement patterns, thus enhancing consistency and reliability of tracking

patterns over time.

4. At the last stage, a live dashboard is developed to track the real-time location data. The system takes away the manual position and security alert automation, which is encrypted and access controlled in the name of privacy. Signal errors and interference are minimized by the use of optimization techniques..

Training

• The system training was initiated by the collection of RSSI values of various reference points known throughout the building in order to develop a fine signal fingerprint map. The data points were marked with the precise position in order to assist the machine learning model to acquire the correlation between signal strength and position.

• These patterns were then trained in the model and enhanced predictions of locations, addressing interference and variations in signal. The accuracy of the same was checked by a different set of test locations. Lastly, the trained model was incorporated into the system to improve real-time positioning and aiding automatic attendance and security measures.

Testing

• System testing was done by initially testing whether the Location Database stored access point coordinates and room details correctly. Various known positions were used to record the RSSI values to check the consistency and stability of signal strength. The gadget used by the professor was linked to the Wi-Fi network to enable it to be accurately registered and identified. Wi-Fi triangulation was used to estimate the location, and it was compared against the physical position to determine the accuracy. The machine learning model was experimented to ensure that it has enhanced precision compared to the basic triangulation outcomes. Lastly, server processing, dashboard display, and automatic attendance or security actions were also put to test so as to make sure the system is running smoothly and reliably.

RESULTS AND DISCUSSION

The system training was initiated by the collection of RSSI values of various reference points known throughout the building in order to develop a fine signal fingerprint map. The data points were marked with the precise position in order to assist the machine learning model to acquire the correlation between signal strength and position.

These patterns were then trained in the model and enhanced predictions of locations, addressing interference and variations in signal. The accuracy of the same was checked by a different set of test locations. Lastly, the trained model was incorporated into the system to improve real-time positioning and aiding automatic attendance and security measures.

Data Loading and Exploration

• The Wi-Fi location data is gathered, cleaned, and stored in one central server and analyzed to determine patterns of movement and plot trends

• The Wi-Fi location data is obtained by scanning the access points within the campus and loaded into the central server upon cleaning and sorting the records.
.

• Processed data is then investigated to determine patterns of movement and illustrate trends in order to understand it..

A. Model Training and Evaluation:

Table 1: Performance Comparison Models

S.no	Model	Accuracy	Precision	Recall	F1 - score
1	Proposed model	94.8	94.5	943.9	94.2
2	KNN	89.6	88.9	88.4	88.6
3	Randomforest	92.3	91.8	91.2	91.4
4	Decision Tree	87.9	87.2	86.8	87.0
5	Neural Network	93.1	92.7	92.0	92.3

Table 1 The table compares the various machine learning models that are applied in Wi-Fi based location prediction in terms of accuracy, precision, recall and F1-score.

Wi-Fi ML model proposed demonstrates the best results in all metrics, which implies higher prediction and reliability.

Random Forest and Neural Network models are good whereas the Decision Tree and KNN are relatively poor.

B. Confusion Matrix and Classification Performance:

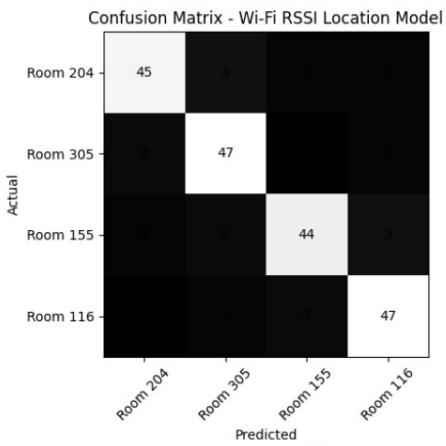

Fig 3: Confusion matrix

Figure 3 This confusion matrix indicates the performance of Wi-Fi RSSI location model in predicting four rooms (204, 305, 155 and 116).

These values are mostly on the diagonal (45, 47, 44, 47) which means that they were correctly predicted in all rooms.

All the small off-diagonal figures indicate different cases of misclassification between the neighboring rooms.

All in all, the model is very accurate with minimal errors in prediction.

C. Performance Training and Validation Accuracy Analysis:

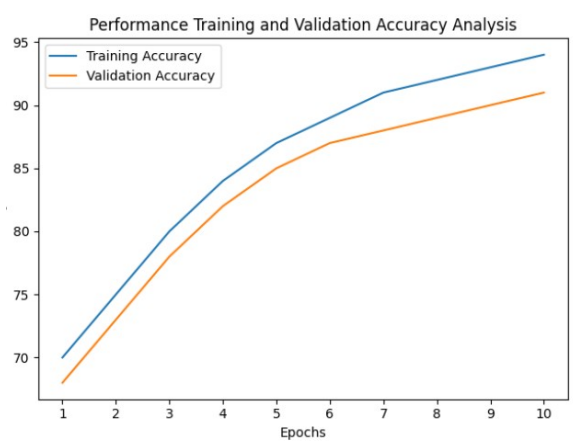

Figure 4. Training and validation accuracy curves

Figure 4 The graph shows the training and validation accuracy of Wi-Fi RSSI location model in the 10 epochs. Both the training and validation accuracy increase steadily as the number of epochs increases and this is a sign of effective learning. Accuracy of the training is increased to 94 percent and this indicates that the model is fitting the training data. Likewise, the accuracy of the validation also rises to 91% demonstrating that the model works well with unknown data. The distance between the training and validation curve is insignificantly small which shows that there is a little overfitting and high generalization ability. The results on the whole testify to the model having stable, reliable, and better performance..

V. **CONCLUSION**:

The graph shows the training and validation accuracy of Wi-Fi RSSI location model in the 10 epochs. Both the training and validation accuracy increase steadily as the number of epochs increases and this is a sign of effective learning. Accuracy of the training is increased to 94 percent and this indicates that the model is fitting the training data. Likewise, the accuracy of the validation also rises to 91% demonstrating that the model works well with unknown data. The distance between the training and validation curve is insignificantly small which shows that there is a little overfitting and high generalization ability. The results on the whole testify to the model having stable, reliable, and better performance.
To summarize, the presented system shows that the geolocation tracking model based on Wi-Fi technology could be successfully used at the university campus to locate and track the objects. Using Wi-Fi RSSI measurements and machine learning algorithms, the system has a high accuracy rate and can ensure the reliable indoor positioning without the use of GPS and other hardware such as Bluetooth beacons.

The implementation provides efficient data processing, real time monitoring, as well as automation of functions including attendance tracking and security alerts. Moreover, data privacy and ethical observance is ensured through the incorporation of encryption, anonymization and controlled access policy. In general, the system based on Wi-Fi offers a good low-cost, scalable, and precise answer to a smart campus location tracking system.

.

VI. **REFERENCES:**

P. Bahl and V. N. Padmanabhan, “RADAR: An in-building RF-based user location and tracking system,” Proceedings of IEEE INFOCOM, [2000].

[1] M. Youssef and A. Agrawala, “The Horus WLAN location determination system,” Proceedings of the 3rd International Conference on Mobile Systems, Applications, and Services (MobiSys), [2005].

[1] K. Kaemarungsi and P. Krishnamurthy, “Modeling of indoor positioning systems based on location fingerprinting,” IEEE INFOCOM, [2004]

[2] H. Liu, H. Darabi, P. Banerjee, and J. Liu, “Survey of wireless indoor positioning techniques and systems,” IEEE Transactions on Systems, Man, and Cybernetics, Vol. 37, No. 6, [2007].

[3] S. He and S.-H. G. Chan, “Wi-Fi fingerprint-based indoor positioning: Recent advances and comparisons,” IEEE Communications Surveys & Tutorials, [2016].

[4] J. Torres-Sospedra et al., “Comprehensive analysis of distance and similarity measures for Wi-Fi fingerprinting indoor positioning systems,” Expert Systems with Applications, [2015].

[5] Y. Wang, X. Jia, and H. K. Lee, “An indoor wireless positioning system based on wireless local area network infrastructure,” Proceedings of IEEE International Conference on Mobile Adhoc and Sensor Systems, [2003].

[6] M. F. Alzantot and M. Youssef, “UPTIME: Ubiquitous pedestrian tracking using mobile phones,” IEEE Wireless Communications and Networking Conference (WCNC), [2012].

[7] S. Mazuelas et al., “Robust indoor positioning provided by real-time RSSI values in unmodified WLAN networks,” IEEE Journal of Selected Topics in Signal Processing, [2009].

[8] A. Haeberlen, E. Flannery, A. Ladd, A. Rudys, D. Wallach, and L. Kavraki, “Practical robust localization over large-scale 802.11 wireless networks,” Proceedings of ACM MobiCom, [2004].